MILITARY SPECIFICATION SHEET

(D) DISPLAY, OPTOELECTRONIC, SEGMENTED READOUT, BACKLIGHTED, STYLE II (LIGHT EMITTING DIODES), RFI SHIELDED, MOISTURE SEALED, HIGH IMPACT SHOCK, TYPE R01

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the displays described herein shall consist of this specification and the latest issue of MIL-D-28803.

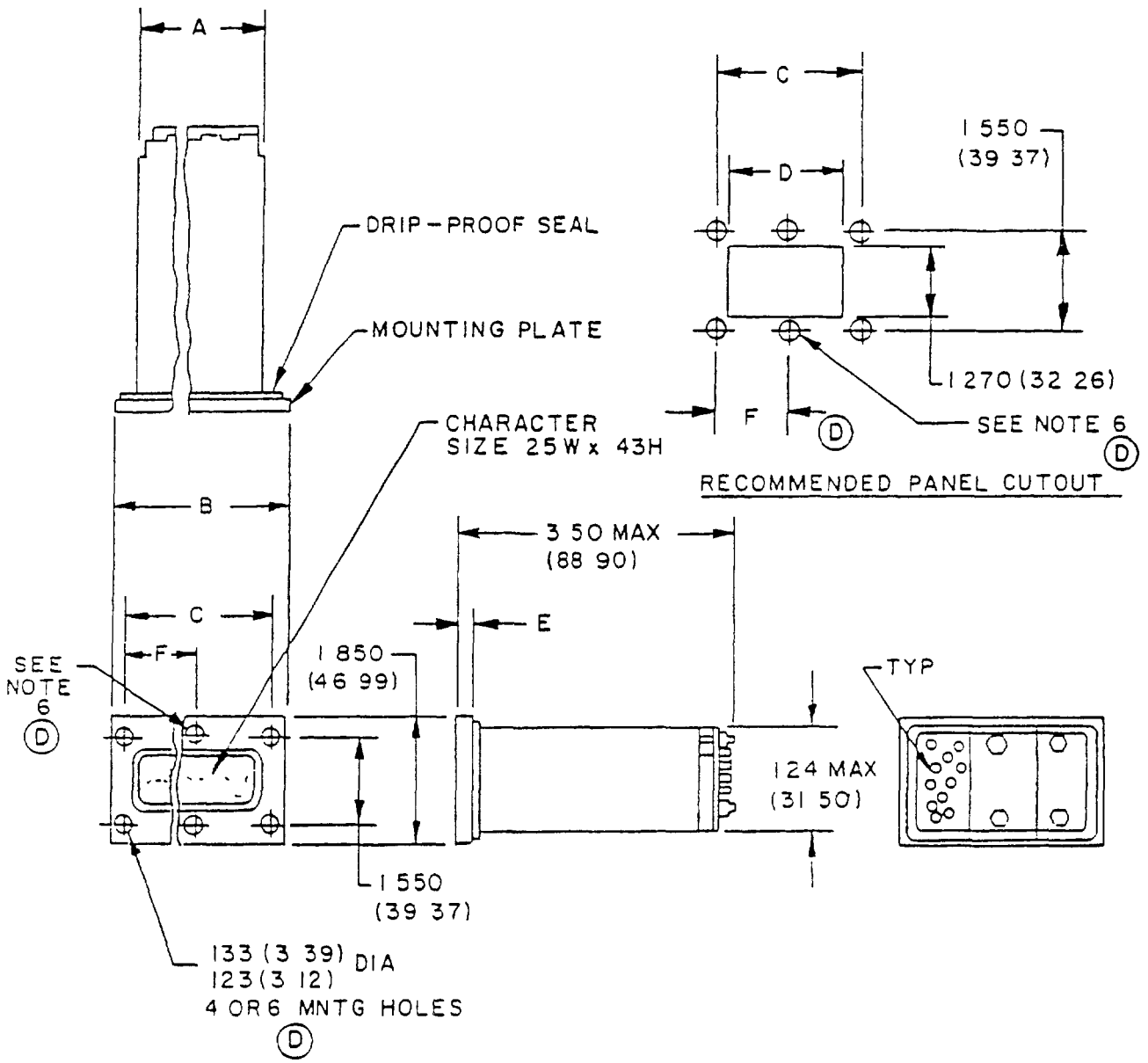


FIGURE 1 Type R01 display and panel cutout

(D) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

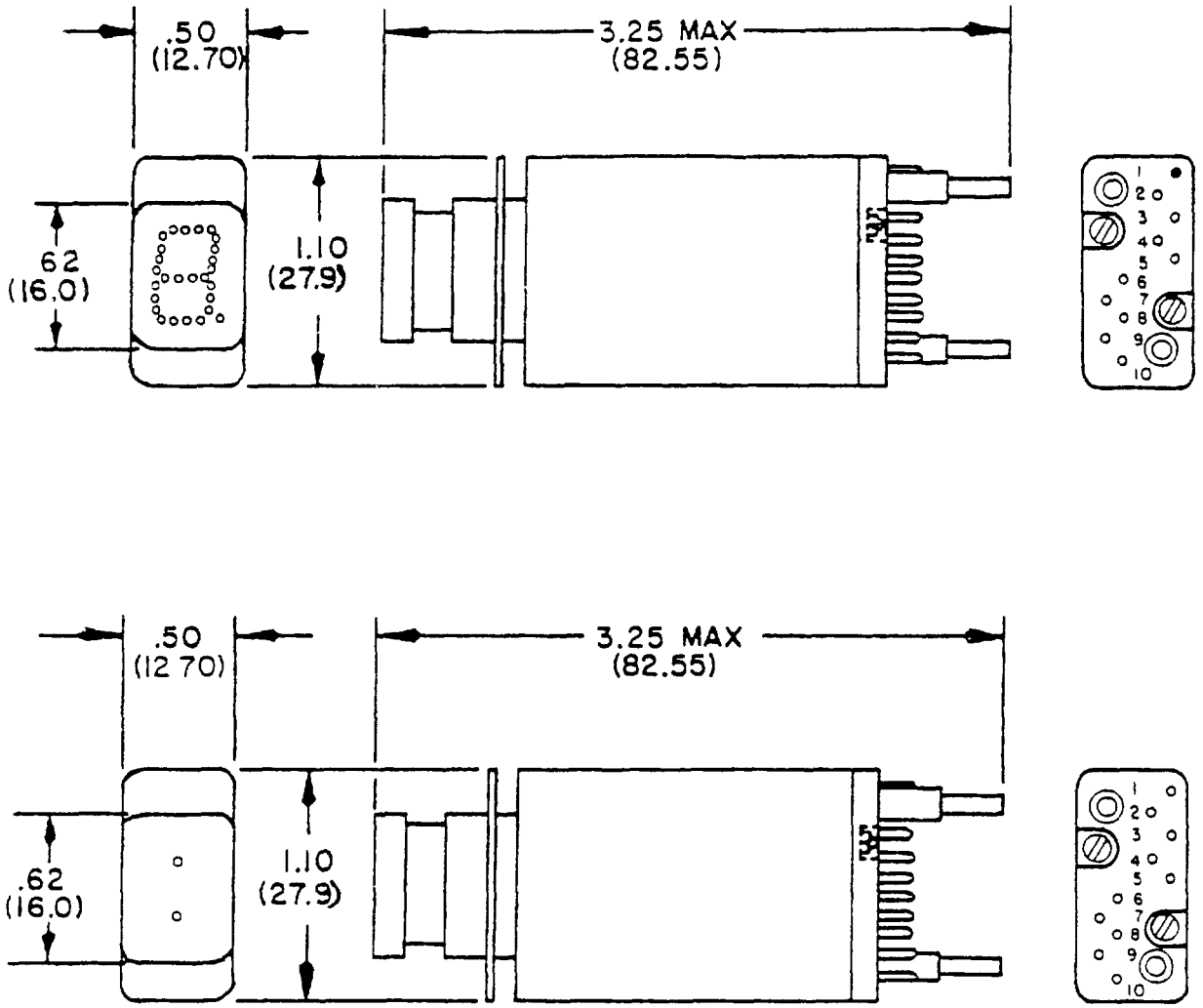
Seal	Dimension E
Dripproof	.22 (5.6)
Immersion	.32 (8.1)

No. of units per assembly	Dimensions					Weight
	A	B	C	D	F immersion seal readouts only	oz max
1	.62 (15.7)	1.187 (30.15)	.867 (22.02)	.67 (17.0)	--	3.5
2	1.12 (28.4)	1.687 (42.85)	1.367 (34.72)	1.17 (29.7)	--	5.0
3	1.62 (41.1)	2.187 (55.55)	1.867 (47.42)	1.67 (42.4)	--	7.0
4	2.12 (53.8)	2.687 (68.25)	2.367 (60.12)	2.17 (55.1)	--	8.5
5	2.62 (66.5)	3.187 (80.95)	2.867 (72.82)	2.67 (67.8)	1.433 (36.40)	10.5
6	3.12 (79.2)	3.687 (93.65)	3.367 (85.52)	3.17 (80.5)	1.683 (42.75)	12.0
7	3.62 (91.9)	4.187 (106.35)	3.867 (98.22)	3.67 (93.2)	1.933 (49.10)	14.0
8	4.12 (104.6)	4.687 (119.05)	4.367 (110.92)	4.17 (105.9)	2.183 (55.45)	16.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional within envelope as shown.
(D) 6. The two center mounting holes (located by dimension F) are required in designated immersion seal readouts only See dimension table herein and table IV.

FIGURE 1. Type R01 display and panel cutout - Continued

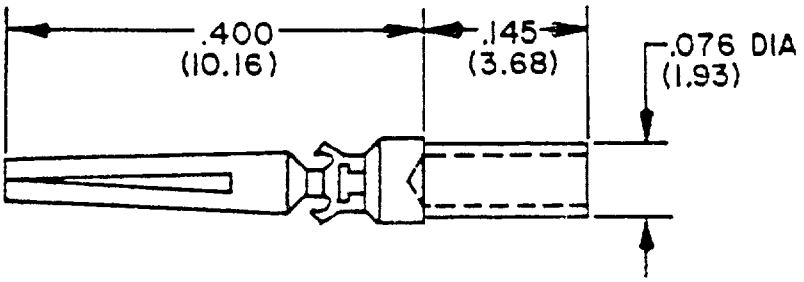
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.



NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerances are ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional within envelope as shown.

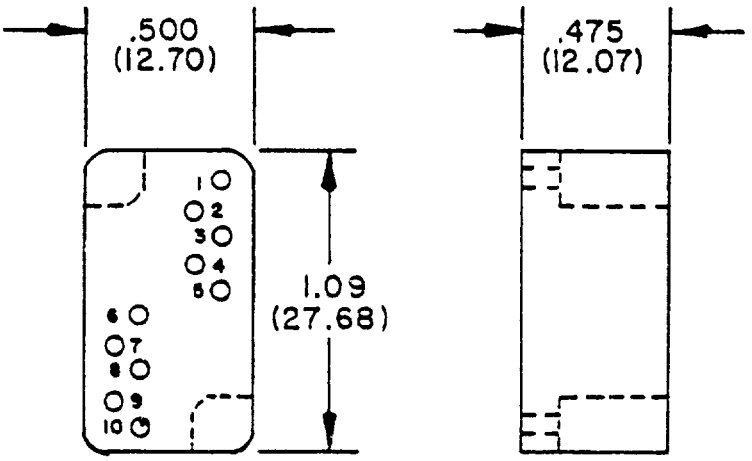
FIGURE 2 Display modules

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals.
5. Design configuration optional within envelope as shown.
6. Terminal part number (package of 25 terminals): M22885/80-50.
(D) 7. Application information:
   Crimp tool - M22520/1-01
   Positioner - M22520/1-02
   Insertion tool - MSC products (CAGE 96182) 800-3191
   Extraction tool - MSC products 800-P7
   Locator tool - MSC products 800-3191-L20

(D) FIGURE 3. Terminal, crimp type.



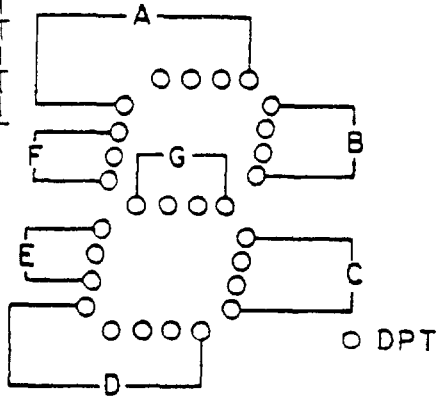
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
5. Design configuration optional within envelope as shown.
6. Connector block part number M28803/1-ZZ.

FIGURE 4. Connector block.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

Truth table															
Circuit configuration 1															
Input							Output								
		BCD					Δ = Segment lit								Display
DPT	EN	8	4	2	1	BI	A	B	C	D	E	F	G	DPT	
	0	0	0	0	0	1	Δ	Δ	Δ	Δ	Δ	Δ			0
	0	0	0	0	1	1		Δ	Δ						1
	0	0	0	1	0	1	Δ	Δ		Δ	Δ		Δ		2
	0	0	0	1	1	1	Δ	Δ	Δ	Δ			Δ		3
	0	0	1	0	0	1		Δ	Δ			Δ	Δ		4
	0	0	1	0	1	1	Δ		Δ	Δ		Δ	Δ		5
	0	0	1	1	0	1			Δ	Δ	Δ	Δ	Δ		6
	0	0	1	1	1	1	Δ	Δ	Δ						7
	0	1	0	0	0	1	Δ	Δ	Δ	Δ	Δ	Δ	Δ		8
	0	1	0	0	1	1	Δ	Δ	Δ			Δ	Δ		9
	0	1	0	1	0	1									Blank
	0	1	0	1	1	1									Blank
	0	1	1	0	0	1									Blank
	0	1	1	0	1	1									Blank
	0	1	1	1	0	1									Blank
	0	1	1	1	1	1									Blank
	X	X	X	X	X	0									Blank
1	X	X	X	X	X	X								Δ 1/	o

1/ The decimal point (DPT) operates independently of the logic. To illuminate, connect to 5.0 V dc.



FRONT VIEW – 7 SEGMENT

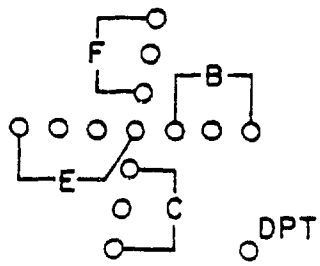
0 = Logic low level, represents ground potential (0.0 V dc).
1 = Logic high level
X = Logical "0" or "1".
DPT = Decimal point.
LS = LED source.
EN = Enable.
BI = Blanking input.

FIGURE 5. Truth table - seven segment numeric display, with latch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

Truth table															
Circuit configuration 2															
Input							Output								
		BCD					Δ = Segment lit								Display
DPT	EN	8	4	2	1	BI	A	B	C	D	E	F	G	DPT	
	X	X	X	X	X	0									Blank
	0	0	0	0	0	1		Δ	Δ		Δ	Δ			+
	0	0	0	0	1	1		Δ			Δ				-
1	X	X	X	X	X									Δ 1/	o

1/ The decimal point (DPT) operates independently of the logic. To illuminate connect to +5.0 V dc.



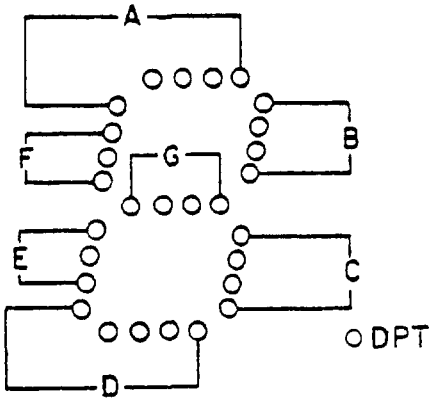
FRONT VIEW – 4 SEGMENT

FIGURE 6. Truth table - four segment sign display, with latch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

Truth table																
Circuit configuration 3																
Input								Display	Δ = Segment lit							
DPT	A	B	C	D	E	F	G		A	B	C	D	E	F	G	DPT
	0	0	0	0	0	0		0	Δ	Δ	Δ	Δ	Δ	Δ		
		0	0					1		Δ	Δ					
	0	0		0	0		0	2	Δ	Δ		Δ	Δ		Δ	
	0	0	0	0			0	3	Δ	Δ	Δ	Δ			Δ	
		0	0			0	0	4		Δ	Δ			Δ	Δ	
	0		0	0		0	0	5	Δ		Δ	Δ		Δ	Δ	
			0	0	0	0	0	6	Δ		Δ	Δ	Δ	Δ	Δ	
	0	0	0					7	Δ	Δ	Δ					
	0	0	0	0	0	0	0	8	Δ	Δ	Δ	Δ	Δ	Δ	Δ	
	0	0	0			0	0	9	Δ	Δ	Δ			Δ	Δ	
0								o								Δ

0 = Logic low level, represents ground potential (0.0 V dc).
DPT = Decimal point.

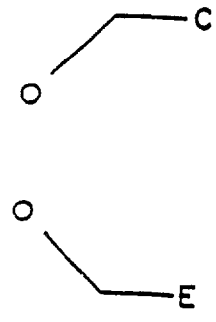


FRONT VIEW – 7 SEGMENT

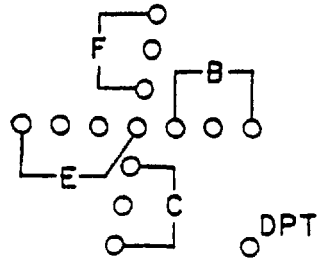
FIGURE 7 Truth table - seven segment direct wired numeric display

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

Truth table										
Circuit configuration 4										
Input					Display	Segment lit				
DPT	B	C	E	F		B	C	E	F	DPT
	0	0	0	0	+	Δ	Δ	Δ	Δ	
	0		0		-	Δ		Δ		
0					o					Δ
		0	0		o o		Δ	Δ		



FRONT VIEW
COLON



FRONT VIEW - 4 SEGMENT

FIGURE 8. Truth table - four segment direct wired sign display and colon.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

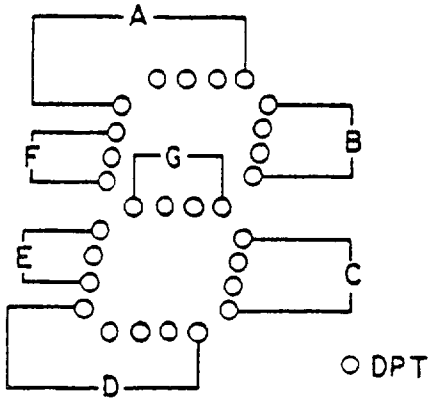
Truth table

Circuit configuration 5

Input							Output								
		BCD					Δ = Segment lit								Display
DPT	LT	8	4	2	1	BI	A	B	C	D	E	F	G	DPT	
	1	0	0	0	0	1	Δ	Δ	Δ	Δ	Δ	Δ			0
	1	0	0	0	1	1		Δ	Δ						1
	1	0	0	1	0	1	Δ	Δ		Δ	Δ		Δ		2
	1	0	0	1	1	1	Δ	Δ	Δ	Δ			Δ		3
	1	0	1	0	0	1		Δ	Δ			Δ	Δ		4
	1	0	1	0	1	1	Δ		Δ	Δ		Δ	Δ		5
	1	0	1	1	0	1			Δ	Δ	Δ	Δ	Δ		6
	1	0	1	1	1	1	Δ	Δ	Δ						7
	1	1	0	0	0	1	Δ	Δ	Δ	Δ	Δ	Δ	Δ		8
	1	1	0	0	1	1	Δ	Δ	Δ			Δ	Δ		9
	1	1	0	1	0	1				Δ	Δ		Δ		
	1	1	0	1	1	1			Δ	Δ			Δ		
	1	1	1	0	0	1		Δ				Δ	Δ		
	1	1	1	0	1	1	Δ			Δ		Δ	Δ		
	1	1	1	1	0	1				Δ	Δ	Δ	Δ		
	1	1	1	1	1	1									Blank
	X	X	X	X	X	0									Blank
0	X	X	X	X	X	X								Δ 1/	o
	0	X	X	X	X	1	Δ	Δ	Δ	Δ	Δ	Δ	Δ		8

1/ The decimal point (DPT) operates independently of the logic. To illuminate, connect to GND (0.0 V dc).

0 = Logic low level, represents ground potential (0.0 V dc).
1 = Logic high level
X = Logical "0" or "1".
DPT = Decimal point.
LT = LED test.
BI = Blanking input.



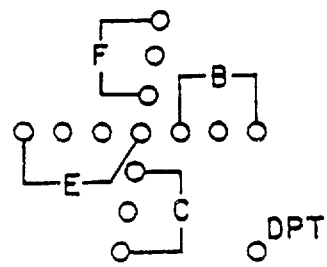
FRONT VIEW - 7 SEGMENT

FIGURE 9. Truth table - seven segment numeric display, without latch

Source: https://assist.dla.mil --Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

Truth table															
Circuit configuration 6															
Input							Output								
		BCD					Δ = Segment lit								Display
DPT	LT	8	4	2	1	BI	A	B	C	D	E	F	G	DPT	
	X	X	X	X	X	0									Blank
	1	0	0	0	0	1		Δ	Δ		Δ	Δ			+
	1	0	0	0	1	1		Δ			Δ				-
0	X	X	X	X	X	X								Δ 1/	o
	0	X	X	X	X	1		Δ	Δ		Δ	Δ			+

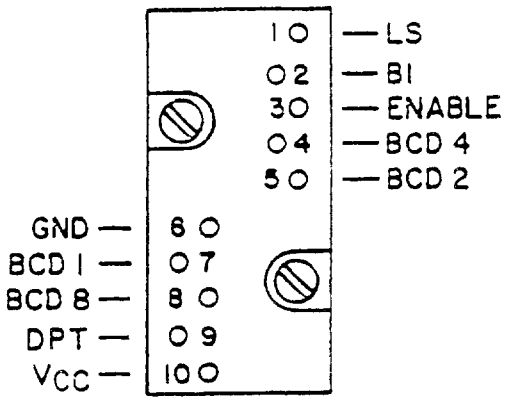
1/ The decimal point (DPT) operates independently of the logic. To illuminate connect to GND (0.0 V dc).



FRONT VIEW – 4 SEGMENT

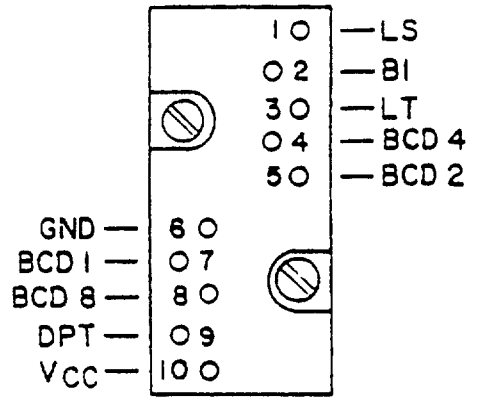
FIGURE 10. Truth table - four segment sign display, without latch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.



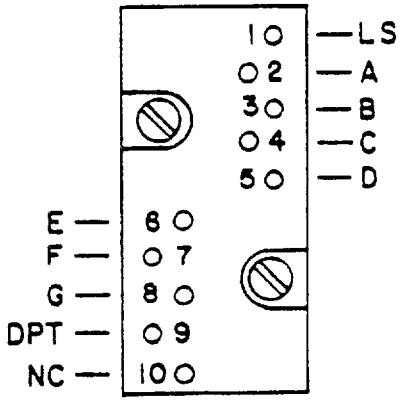
WITH LATCH

FOR CIRCUIT CONFIGURATIONS 1 AND 2



WITHOUT LATCH

FOR CIRCUIT CONFIGURATIONS 5 AND 6



DIRECT WIRED

FOR CIRCUIT CONFIGURATIONS 3 AND 4

VCC = SUPPLY VOLTAGE
GND = GROUND
DPT = DECIMAL POINT
NC = NO CONNECTION
LS = LED SOURCE
LT = LED TEST
BI = BLANKING INPUT

FIGURE 11. Terminal designation (rear view).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration. See figures 1 through 4.

(D) Display assembly. The display assembly shall consist of the following:

a. Multistation matrix frame.

(D) b. Display module and circuit assembly.

c. Insertable solder-type terminals (packaged separately).

d. RFI screen.

e. Light emitting diodes.

f. Terminal block.

(D) NOTE Encoders used in the fabrication of displays shall be screened in accordance with the class B requirements of method 5004 of MIL-STD-883 and tested in accordance with the class B requirements of method 5005 of MIL-STD-883.

Matrix frame: Includes connector blocks, fasteners, and front lens plate.

Front lens plate: Shall appear obscured in the unlighted condition. In the lighted condition, characters shall appear in the color specified on a black background.

RFI shielding: Grounded lens screen is provided.

Mounting flange: Front face of mounting flange shall be a black, lusterless color.

Temperature characteristic: -55°C to +85°C.

Electrical:

Circuit characteristics: See figures 5 through 11 and tables I and II.

(D) Leakage (immersion): MIL-STD-810, method 512, procedure I. Following the test, displays shall meet the dielectric withstanding voltage requirements and there shall be no evidence of leakage.

Vibration grade: 2.

Shock: Method II (high impact).

Terminal strength: 5 pounds, test condition letter A.

Chromaticity: Color red, method II (Y = .335 maximum, Z = 002 maximum).

Dot-to-dot ratio (per character): 5 to 1 maximum after life (electrical).

Contrast ratio: 20 to 1 minimum at 110 foot-candles incidental illumination.

Code letters for combinations. See tables III and IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

TABLE I Electrical specifications for configurations 1 and 2

Symbol	Characteristic	Limits			Unit	Conditions
		Min	Typ	Max		
V_{CC} 1/	Supply voltage	4 75	5 00	5 25	V	T_A = +25°C
V_{IL}	Logical "0" low input voltage			1 50	V	V_{CC} = 5 V
V_{IH}	Logical "1" high input voltage	3 50			V	V_{CC} = 5 V
I_{IN}	Input current		10		pA	
V_{LS}	LED supply voltage	2 5		0	V	1/ 2/ 3/
I_{LS}	LED supply current			200	mA	V_{CC} = 5 V
t_{setup}	Setup time		90		ns	
t_{hold}	Hold time		90		ns	V_{CC} = 5 V
PW_{LE}	Minimum latch enable pulse width		260	520	ns	V_{CC} = 5 V

1/ Device should not be connected with power on.

2/ GND (0 0 V dc) is maximum intensity, increasing the voltage will decrease intensity

3/ Use of LED supply to regulate brightness is not recommended, as display uniformity cannot be maintained. Pulse width modulation of the blanking input (BI) is the preferred method

Source: https://assist.dla.mil - Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical specifications for configurations 5 and 6.

Symbol	Characteristic	Limits			Unit	Conditions
		Min	Typ	Max		
V_{CC}	Supply voltage	4.75	5.0	5.25	V	T_A = +25°C
V_{IH}	Input high voltage	2.0			V	Guaranteed input high voltage for all inputs
V_{IL}	Input low voltage			0.8	V	Guaranteed input low voltage for all inputs
I_{IH}	Input high current			40	uA	Logical "1" level input current at any input except BI/RBO. V_{CC} = maximum, V_{IN} = 2.4 V
I_{IL}	Input low current			-1.6	mA	Logical "0" level input current at any input except BI/RBO. V_{CC} = maximum, V_{IN} = 0.4 V
				-4.0	mA	Logical "0" level input current at BI/RBO. V_{CC} = maximum, V_{IN} = 0.4 V
I_{CC}	Power supply current		64	85	mA	V_{CC} = maximum less LED's (D)
V_{LS}	LED supply voltage	2.5		5.0	V	1/ 2/ 3/
I_{LS}	LED supply current			200	mA	V_{CC} = maximum

1/ Device should not be connected with power on.

2/ Use of LED supply to regulate brightness is not recommended, as display uniformity cannot be maintained. Pulse width modulation of the blanking input (BI) is the preferred method.

3/ +5.0 V dc is maximum intensity. Decreasing the voltage will decrease intensity.

Source: https://assist.dla.mil Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

TABLE III. Code letters for combinations of number of stations and type and arrangement of modules

Code letter	Number of modules	Type of module and arrangement
A	Module only 1/	4 segment
B	Module only 1/	7 segment
C	1 module assembly	4 segment
D	1 module assembly	7 segment
E	2 module assembly	All 7 segment modules
F	3 module assembly	All 7 segment modules
G	4 module assembly	All 7 segment modules
H	5 module assembly	All 7 segment modules
J	6 module assembly	All 7 segment modules
K	7 module assembly	All 7 segment modules
L	8 module assembly	All 7 segment modules
M	2 module assembly	1 4 segment followed by 1 7 segment
N	3 module assembly	1 4 segment followed by 2 7 segment
P	4 module assembly	1 4 segment followed by 3 7 segment
Q	5 module assembly	1 4 segment followed by 4 7 segment
R	6 module assembly	1 4 segment followed by 5 7 segment
S	7 module assembly	1 4 segment followed by 6 7 segment
T	8 module assembly	1 4 segment followed by 7 7 segment
U	5 module assembly	2 7 segment, 1 colon, 2 7 segment
V	8 module assembly	2 7 segment; 1 colon, 2 7 segment, 1 colon; 2 7 segment
W	Module only 1/	Colon

1/ Insertable crimp type terminals and connector blocks are to be furnished with modules

TABLE IV. Code letters for combinations of type of circuitry and seal.

Code letter	Type of circuitry	Seal
A	Direct wired	Dripproof
B 1/	Conversion logic without latching	"
C	Conversion logic with latching	"
D	Direct wired	Immersion
E 1/	Conversion logic without latching	"
F	Conversion logic with latching	"

1/ Not available with colon module. Colon portion of assembly supplied with direct wired circuitry only.

Source: https://assist.dla.mil Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.

QUALIFICATION:

Extent of qualification: See table V.

TABLE V. Extent of qualification.

Dash number	Number of samples required	Tests	Qualifies
GA GB PF	4 4 1/ 4	Complete in accordance with qualification inspection of MIL-D-28803 2/	All

1/ 2 sample units for group II tests, 1 sample unit for group III tests, 1 sample unit for group IV tests.
2/ When applicable, leakage (immersion) test shall be performed in lieu of dripproof seal test.

Part number: The part number shall consist of the letter "M" followed by the basic number of the specification sheet, and a coded two digit dash number indicating the number, type, and arrangement of the modules, and type of circuitry and seal, as specifed. Example.

M28803/1- M C

Military designator and specification sheet number

Single code indicating number, type, and arrangement of modules, as specified (see table III)

Single code indicating seal and type of circuitry, as specified (see table IV)

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AT, AV, ME, MI
Navy - MC, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5980-0003)

Source: https://assist.dla.mil Downloaded: 2026-10-20T18:57Z
Check the source to verify that this is the current version before use.